$Factorial - Growth\ Rate\ of\ Output$

vs

$Time\ Complexity$

$Growth\ Rate\ of\ Output$:

```
int factorial(int n)
{
    // Base case: if n is 0 or 1, return 1
    // This is because the factorial of 0 and 1 is 1
    if (n == 0 || n == 1)
    {
        return 1;
    }
    else
    {
        // Recursive case: n times factorial of n-1
        // This is the definition of factorial: the product
of an integer and all the integers below it
        return n * factorial(n - 1);
    }
}
```

$\underline{Growth\ Rate\ of\ Output}$

$Here\ we\ see$:

$Fact(n) = n \times Fact(n-1)$

$We\ know\ , Fact(0) = 1\ and\ Fact(1) = 1\ hence\ both\ are\ constants\ i.e.`C`\ ,$

$we\ can\ use\ `1\ or\ C`\ just\ to\ represent\ constant. Not\ from\ Time\ Complexity\ but$

$like\ Time\ Complexity\ .$

$For, Fact(n-1), we\ get$:

$Fact(n-1) = n-1\ \times Fact(n-2)$

$Subtituting\ this\ for\ Fact(n-1), we\ get$:

$Fact(n) = n\ \times n-1\ \times Fact(n-2)$

$For, Fact(n-2), we\ get$:

$Fact(n-2) = n-2\ \times Fact(n-3)$

$Subtituting\ this\ for\ Fact(n-2), we\ get$:

$Fact(n) = n\ \times n-1\ \times n-2\ \times Fact(n-3)$

$Therefore\ if\ it\ runs\ for\ `k`\ times\ , then\ we\ get$:

$Fact(n) = n\ \times n-k\ \times n-k+1\ \times \ldots \times n-1\ \times n-2\ \times Fact(n-k)$

$Rearranging\ or\ Resequencing\ we\ get$:

$Fact(n) = n-k\ \times n-k+1\ \times \ldots \times n-2\ \times n-1\ \times n \times Fact(n-k)$

$Now, say\ , n-k = 1\ \ , then, k = n-1, there\ substituting\ this\ in\ `k`\ we\ \ get$:

$Fact(n) = n-(n-1)\ \ \times n-(n-1)+1\ \times \ldots \times n-2\ \times n-1\ \times n \times Fact\big(n-(n-1)\big)$

$Fact(n) = n-n+1\ \ \times n-n+1+1\ \times \ldots \times n-2\ \times n-1\ \times n \times Fact(n-n+1)$

$Fact(n) = 1\ \ \times 2\ \times \ldots \times n-2\ \times n-1\ \times n \times Fact(1)$

$Fact(n) = 1\ \ \times 2\ \times \ldots \times n-2\ \times n-1\ \times n \times 1\ [Fact(1) = C\ or\ `1`\ i.e. Constant\ ]$

$\therefore Fact(n) = 1\ \ \times 2\ \times \ldots \times n-2\ \times n-1\ \times n$

$\Rightarrow \boldsymbol{Fact(n) = n!}$

$\therefore$ ***Growth Rate of Output of Factorial*** $= \mathbf{O}(\boldsymbol{n}!)$ ***, while ,***

Time Complexity of Factorial

$$T(n) = \underbrace{T(n-1)}_{\boxed{\textit{To Compute } F(n-1)}} + \underbrace{1}_{\boxed{\textit{To multiply } F(n-1) \textit{ by } n}} \quad , \boldsymbol{for\ n > 0}$$

$\Longrightarrow \boldsymbol{O(n)}$

i.e. Growth Rate of Output $= \boldsymbol{n!}$ ***but Time Complexity of the Algorithm remains at `n`*** .